Create probability matrices

documents / words

d_1: the^{1} brown^{2} fox^{3}

d_2: for^{3} jumps^{4} over^{5} the^{1} moon^{6}

d_3: moon^{6}

$$f = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{array}$$

Then $F_j = \sum_{i=1}^{6} f_{ij} = \begin{cases} F_1 = 3 \\ F_2 = 5 \\ F_3 = 1 \end{cases} \rightarrow p_{ij} = \frac{f_{ij}}{F_j} \rightarrow P = \begin{bmatrix} 1/3 & 1/5 & 0 \\ 1/3 & 0 & 0 \\ 1/3 & 1/5 & 0 \\ 0 & 1/5 & 0 \\ 0 & 1/5 & 0 \\ 0 & 1/5 & 1 \end{bmatrix}$

→ weighted

Then if $P_j^w = P_j \cdot w_j$, we can define $w_j = \log(1 + |u(d_j)|)$, with $u(d_j)$ = qty of upvotes in d_j, so as to not influence edges too much (layout!!).

Then if e.g. $u(d_1) = 1$, $u(d_2) = 3$ and $u(d_3) = -9$, we get:

$$w = \begin{bmatrix} \log(2) \\ \log(4) \\ 1 \end{bmatrix} \rightarrow P_j^w = P_j \cdot w_j \rightarrow j = 1 \rightarrow P_1^w = P_1 \cdot w_1 = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot \log(2)$$

* we want big number of upvotes, regardless of sign, to be influential, since they spark controversy (and so that the logarithm is defined $\forall u(d_j)$).

By having P^w, which is formed by columns P^w_j, we can normalize it:

$$p^w_{norm} = \frac{p^w}{\sum_i \log(1+|u(d_i)|)}$$

Then $0 \leq (P^w_{norm})_{ij} \leq 1$ by definition!

We will refer to P^w_{norm} as P_{wd} from now on.

Notice how this essentially identifies a bipartite graph.

Semantic Network (words)

To obtain the semantic network relating words to words, we can use the projection on words of P_{wd} to obtain P_{ww}, which is the adjacency matrix of the desired network.

$$P_{ww} = P_{wd} \cdot diag(p_d)^{-1} \cdot P_{wd}^T \qquad p_d = P_{wd}^T \cdot 1$$

We can analyze this network using Gephi, by calculating:

- degree (power law?) to find out how important the nodes are;
- PageRank to find out hubs (words used in conjuction with authorities) and authorities (most used words);
- clustering coefficient to find out the graph's tendency to being organized into clusters;

- modularity for clustering and visualization.
- ?

Semantic Network (documents)

To obtain the semantic network relating documents to documents, we can use the projection on P_{wd} to obtain P_{dd}, which is the adjacency matrix of the desired network.

$$P_{dd} = P_{wd}^{T} \cdot diag(p_w) \cdot P_{wd} \qquad p_w = P_{wd} \cdot 1$$

We can analyze this network using Gephi to calculate:

- degree (power law) to find out the importance of nodes;
- PageRank to find out hubs (marginally important documents) and authorities (most important documents in the discussion.

Then, we can use this for topic detection using Louvain to try and highlight relevant topics between documents in Python. We can also use BERTopic to compare it with Louvain using:

- modularity;
- infomap;
- NMI;

- # topics;
- Nmut.

Based on these metrics, we decide which of the two algorithms we use, and consequently which topics to use.

Emotion analysis

We can apply emotion analysis to various datasets, mainly:
- whole dataset;
- programmers dataset;
- designers dataset;
- on each topic found previously.

For each of these, we shall plot a line graph showing the evolution over time (Q) of the chosen emotions:

anger, pessimism, fear, joy, optimism, trust

We can then compare the various obtained results between topics and, more generally, the two communities.

Sentiment analysis

We can apply sentiment analysis to the same datasets defined for emotion analysis.

For each of these, we can define the net sentiment as:

net sentiment = positive score − negative score

Then, we can compute the mean net sentiment and the mean of the upvotes over time (Q) and plot it.

Then, to show how meaningful the means are, we can show the quarterly number of comments.

We can then compare the various obtained results between topics and, more generally, the two communities.

[...]